SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I shalinee KG(192511048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Graph Search Algorithms: BFS, DFS and 8-Puzzle

Question 1: Breadth-First Search (BFS)

Given graph: A is the root node, B and C are its children, and D, E, F, G are the next-level nodes.

1. BFS Traversal Table

Iteration	Queue (after removing the visited node)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

2. BFS Traversal Order

A → B → C → D → E → F → G

3. Queue Contents After Each Iteration

- Iteration 1: A is visited; Queue = [B, C]
- Iteration 2: B is visited; Queue = [C, D, E]
- Iteration 3: C is visited; Queue = [D, E, F, G]
- Iteration 4: D is visited; Queue = [E, F, G]
- Iteration 5: E is visited; Queue = [F, G]
- Iteration 6: F is visited; Queue = [G]
- Iteration 7: G is visited; Queue = [ ]

Conclusion: BFS visits nodes level by level, from left to right, using a FIFO (First In, First Out) queue.

Question 2: Depth-First Search (DFS)

Starting from node A, traverse the left child first.

Completed Table

Iteration	Stack (after visiting)	Visited Node
1	B, C	A
2	D, E, C	B
3	E, C	D
4	C	E
5	F, G	C
6	G	F
7	Empty	G

DFS Traversal Order: A → B → D → E → C → F → G

Note: The stack is shown with the top at the left. The table assumes left-to-right traversal by pushing the right child before the left child.

Question 3: 8-Puzzle Using Breadth-First Search (BFS)

Initial State

1	3	6
5	□	2
4	7	8

Goal State

1	2	3
4	5	6
7	8	□

Solvability check: Reading the initial state row by row and ignoring the blank gives 1, 3, 6, 5, 2, 4, 7, 8. The inversions are (3,2), (6,5), (6,2), (6,4), (5,2), (5,4), which is 6 in total. The goal state has 0 inversions. Both are even, so they have the same parity and the puzzle is solvable.

Answers to the Questions

1. Which node is expanded in each iteration?

BFS expands the initial state first, then each generated state in FIFO queue order, level by level. The blank tile moves in the order Up, Down, Left, Right, and states already generated are not added again. The solution is found at depth 8; with this move order, BFS expands 311 states before it removes the goal from the queue.

2. How many states are generated in total?

With the move order above, 500 distinct states are generated (including the initial state) by the time the goal is reached. For reference, the complete reachable state space of the 8-puzzle contains 181,440 states, which BFS would only exhaust if the goal were unreachable.

3. Write the complete solution path from the initial state to the goal state.

The shortest solution has 8 moves (moving the blank tile): Right, Up, Left, Down, Left, Down, Right, Right.

Step	Row 1	Row 2	Row 3
Initial	1 3 6	5 □ 2	4 7 8
1. Right	1 3 6	5 2 □	4 7 8
2. Up	1 3 □	5 2 6	4 7 8
3. Left	1 □ 3	5 2 6	4 7 8
4. Down	1 2 3	5 □ 6	4 7 8
5. Left	1 2 3	□ 5 6	4 7 8
6. Down	1 2 3	4 5 6	□ 7 8
7. Right	1 2 3	4 5 6	7 □ 8
8. Right (Goal)	1 2 3	4 5 6	7 8 □

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores states level by level. Since every move has equal cost, the first time it reaches the goal, it has found the solution with the minimum number of moves.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.